upGrad

Methodology Report:
Visualisation & Analysis on Namma Yatri Data

Include your visualisations, analysis, results, insights, and outcomes.

Explain your methodology and approach to the tasks. Add your conclusions to the sections.

Table 1: Data Description

Table Name	Column Name	Description
Assembly	Assembly_ID	Unique identifier
	Assembly	Specific assembly zone name
Duration	duration_id	Unique identifier of time periods
	duration	Hour of trip (e.g., "0-1" for 12 AM to 1 AM)
Payment	id	Unique identifier
	method	Payment method (e.g., Cash, UPI, Credit Card)
Trip Details	tripid	Unique identifier of trips
	loc_from	Source Location code
	searches	Trip request count
	searches_got_estimate	Got an estimated price (1 = user gets an estimate, 0 = does not get an estimate)
	searches_for_quotes	Searched for drivers after estimate (1 - searched, 0 - not searched)
	searches_got_quotes	Got quotes (1 = Driver allotted, 0 = not allotted)
	customer_not_cancelled	Whether customer cancelled or not (1 = Not cancelled)
	driver_not_cancelled	Whether driver cancelled or not (1 = Not cancelled)
	otp_entered	(1 = OTP entered, 0 = not entered)
	end_ride	Whether ride was completed (1 = Completed)
Trips	tripid	Links to Trip Details
	faremethod	Payment method ID, links to Payment table
	fare	Fare amount
	loc_from	Location ID of source
	loc_to	Location ID of destination, links to Assembly table
	driverid	Driver ID
	custid	Customer ID
	distance	Distance in KM from source to destination
	duration	Unique identifier of time periods like duration_id

DO NOT EDIT ANYTHING ON THIS PAGE

Points to Note:

1. **Without this methodology document, the other parts of your case study will not be evaluated.**
2. This assignment is different from the ones you have solved before. Make sure that you treat this case study as a storytelling exercise and not an analysis/visualisation one. This will help you be better prepared for the presentations.
3. Once you are done with the analysis and visualisations, there will be many insights at your hand.
   Make sure that you map the right visuals and takeaways with the right audience since some of these insights might be relevant to one group but not to the other group.
4. **DO NOT** change the text or numbering of any task, as it may cause problems with grading. Write your solutions to a task in the space provided below the respective task.

Tasks to be performed

- Present the overall approach of the analysis.
- Mention the problem statement and the analysis approach briefly.
- To solve a task, you have to create relevant visualisations and derive appropriate insights from the visualisations.
- Add all the plots, insights, calculated field commands, results and outcomes for a task with proper numbering and sequence in the report.
- The scores for all tasks (except conclusions) comprise both analysis work in the visualisation tool and its outcome in the report.
- You will be awarded a score for a task only if the Tableau/PowerBI analysis is correct and is included in the report along with the subsequent insights.
- Finally, draw conclusions based on the analysis.

Scoring:

Report Total Marks: 70

Sections: 3 sections (10 marks + 40 marks + 20 marks)

Analysis and Visualisation

1. Data Preparation [10 Marks]

1.1. Import and Join Tables Correctly [5 Mark]

- Import the Namma Yatri dataset into Tableau/Power BI.
- Ensure that you correctly join all tables to create a unified dataset for analysis.
- Verify the relationships between different tables and confirm that data from various sources is properly aligned for accurate insights.

Solution:

<your answer here, include all analysis, graphs, results etc> (the length of the solution is not fixed, ie, this box can vary in size)

Methodology & Approach

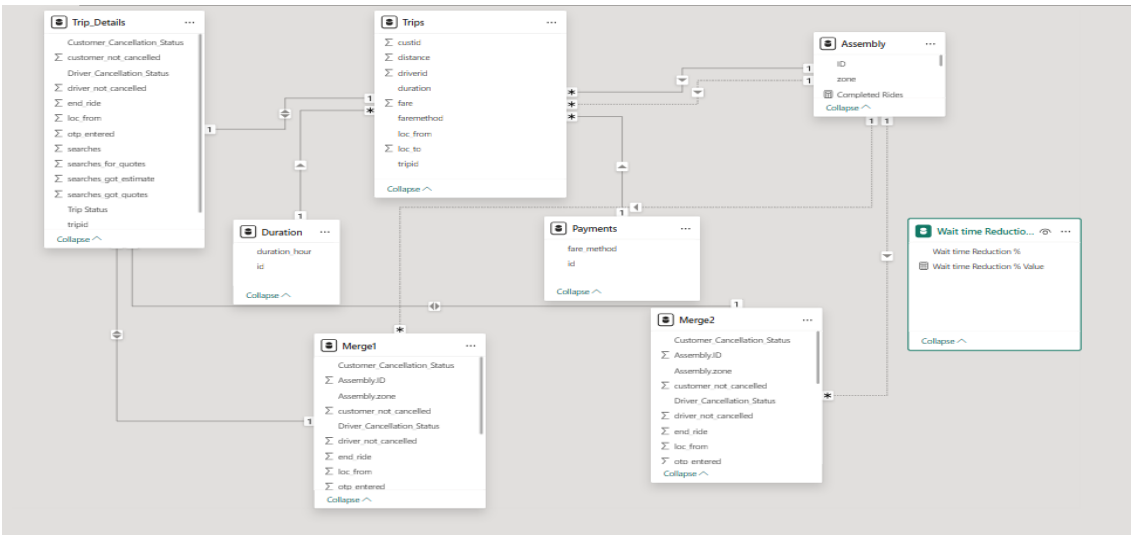
The approach is to import all raw data tables (Trips, Trip_Details, Payment, Duration, Assembly) and establish relationships to create a **star-like schema**. The primary and foreign keys are then identified to create the relationships. This model consolidates attributes, which simplifies the creation of visualizations and measures.

Analysis / Results

The resulting data model places core trip data at the center, with dimension tables (Payment, Duration, Assembly) linked around it. These relationships enable the direct calculation of key performance indicators (KPIs) and allow for a unified analysis.

Key Relationships

- **Trips** links to **Trip Details** on tripid.
- **Trips** links to **Payment** on faremethod and id.
- **Trips** links to **Duration** on duration and id.
- **Trips** links to **Assembly** on loc_from and loc_to to Assembly_ID.



1.2. Find and Resolve Inconsistencies [5 Marks]

- Identify and resolve any inconsistencies or issues in the dataset that might affect the analysis.
- Clean the data to ensure it is structured properly for analysis, removing any irrelevant, duplicate, or erroneous entries.
- While performing the analysis, create calculated fields as needed to ensure the accuracy and relevance of the insights.

Solution:

Incorrect Data Types: Ensured Required columns are set to a numerical data type

Duplicates: Checked for duplicate rows in trips- `tripid`. This will ensure that each trip has a unique tripid to prevent inflated counts

<u>Analysis:</u>

Data preparation involved cleaning duplicates and handling missing values across all categorical fields. We then **converted binary transaction flags** (`'end_ride'`, `'customer_not_cancelled'`, etc.) into specific **calculated measures** for successful rides, customer cancellations, and driver cancellations, Total Quotes , Total Revenue ensuring **analytic consistency**.

2. Exploratory Data Analysis [40 Marks]

2.1. Classify Variables into Categorical and Numerical [2 Marks]

- Classify all the variables in the dataset into numerical and categorical types.

Solution:

Numerical Variables

Based on the data description, the following variables are classified as numerical, as they represent a measurable quantity:

- `fare`: Fare amount.
- `distance`: Distance in KM from source to destination.
- `searches`: Trip request count.
- `customer_not_cancelled`: Indicates if the customer did not cancel (1 or 0).
- `driver_not_cancelled`: Indicates if the driver did not cancel (1 or 0).
- `loc_from`: Source Location code.
- `loc_to`: Location ID of destination
- `searches_got_estimate`: Indicates if a user got an estimated price (1 or 0).
- `searches_for_quotes`: Indicates if a user searched for drivers after an estimate (1 or 0).
- `searches_got_quotes`: Indicates if quotes were received (1 or 0).
- `otp_entered`: Indicates if an OTP was entered (1 or 0).
- `end_ride`: Indicates if a ride was completed (1 or 0).
- `tripid`: Unique identifier for trips.
- `driverid`: Driver ID.
- `custid`: Customer ID.
- `id` (from Payment table): Unique identifier.

Categorical Variables

- `Assembly_ID`: Unique identifier for an assembly zone.
- `Zone` : Specific assembly zone name.
- `method`: Payment method (e.g., Cash, UPI, Credit Card).
- `faremethod`: Payment method ID.

INSIGHT- Through this classification is the foundational step for all further analysis. It determines how variables can be used in visualizations and calculations.example- fare and distance can be used for the will be used to group and filter the data to

identify trends and patterns. This enables the creation of a clear and logical data model.

2.2. Analyse Ride Demand Over Time [3 Marks]

- Explore the distribution of ride demand over time, including trends across different periods.
- Identify the peak demand periods. Choose an appropriate parameter for demand based on your own understanding.

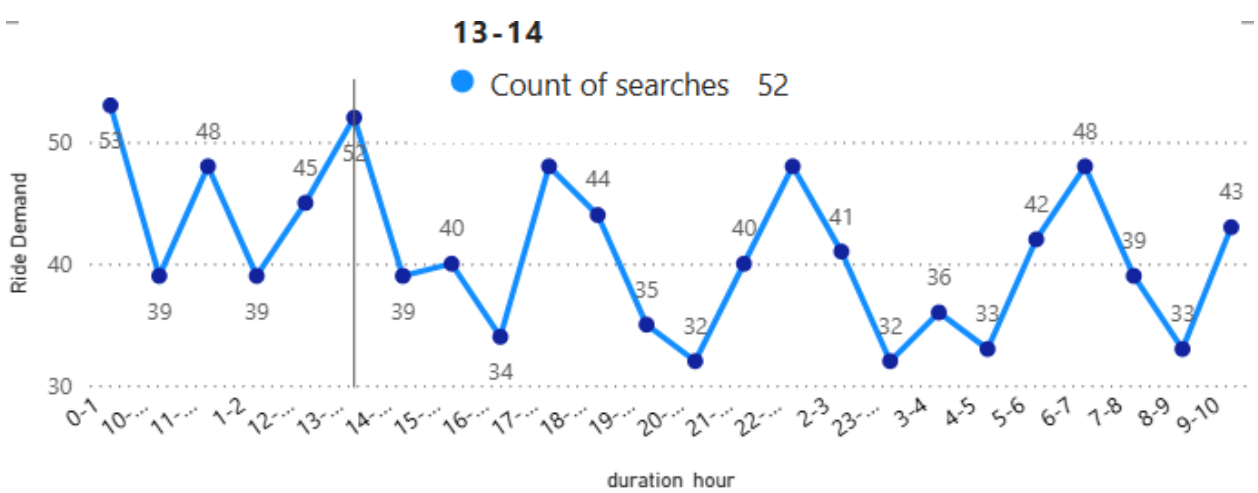
Solution:

Treating blank colum in duration

Analysis-

- Morning Peak: The first major peak in ride requests occurs during the morning commute hours, specifically between 12 AM and 1 AM.
- Evening Peak: The second and notably higher peak is observed during the evening hours, from 2 PM to 3 PM, corresponding to the end-of-day commute.

Visualisation-



Insights-

These two periods represent the peak demand periods for the Namma Yatri service.

By identifying the peak demand periods, the company can strategically allocate its resources, ensuring a higher number of drivers are available during these specific hours to meet customer demand. This proactive approach can help minimize wait

times, reduce potential cancellations, and ultimately improve customer satisfaction and service efficiency

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2.3. Proportion of Total Revenue from Different Time Periods [3 Marks]

- Calculate the proportion of revenue generated during different time periods and visualise how it contributes to total revenue.

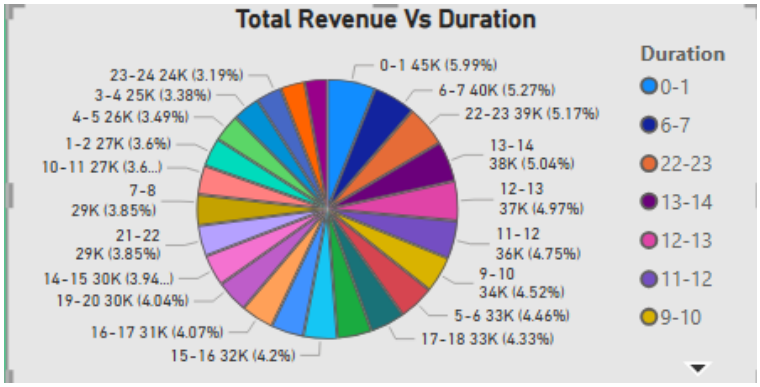
Solution:

Methodology & Approach-

To calculate and visualize the proportion of revenue, the approach is to first create a measure for the total revenue and then use a chart that is effective at showing parts of a whole. The fare column is used to calculate revenue, and the duration column from the Duration table is used to categorize the data by time period.

Visualisation-

A pie chart is created



- Legend: `duration` (from the `Duration` table)
- Values: A DAX measure for `Total Revenue`

The chart shows how each hourly period's revenue contributes to the overall total.

Analysis -

The pie chart reveals that the proportion of total revenue is not evenly distributed across all time periods. The analysis shows that the periods with the highest demand also generate the largest share of the total revenue.

- **I have plot pie-chart which showing Highest Revenue(45,019) in the Duration(0-1).**

Insights-

It provides a clear understanding of the company's financial dynamics. The primary insight is that Namma Yatri's profitability is highly concentrated during the morning .This information is critical for business strategy, as it can be used to:

- Implement targeted surge pricing during peak periods to maximize revenue.
- Offer driver incentives during these specific hours to ensure high service availability.
- Focus marketing efforts on commuters and other users who are most likely to use the service during these profitable time slots.

2.4.Explore the Relationship Between Trip Hour and Revenue [3 Marks]

- Investigate the correlation between trip hour and total fare.
- Explain any trends or patterns that emerge.

Solution:

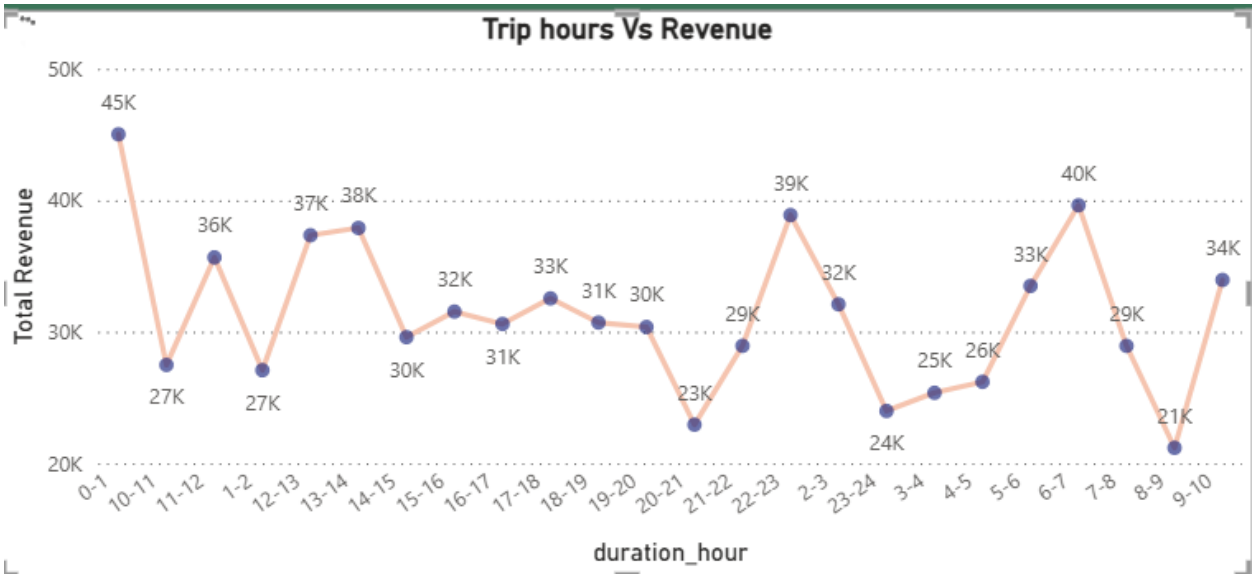
Methodology -

- The analysis used a Line Chart to visualize the trend of revenue over the 24-hour day.
- A measure for Total Revenue was calculated by summing the fare column across all completed trips.

Approach-

- Analysis investigates the relationship between the `Trip hour` and the `Total Revenue` generated from completed trips to identify high-value time slots.

Visualisation-



- **X-axis:** The `duration` column.
- **Y-axis:** The `Total Revenue` measure.

This chart visually represents how revenue fluctuates throughout the day.

Analysis –

The analysis reveals a strong positive correlation between the trip hour and total fare. The line graph shows that total revenue peaks during the same hours as the ride demand, which were identified as the morning **(12-1 AM) and (6-7 AM) rush hours.**

Insights-

The primary insight is that revenue is not evenly distributed but is highly concentrated during the busiest periods of the day.

2.5. Examine the Popularity of Different Payment Methods [3 Marks]

- Analyse the distribution of various payment methods used by customers.
- Identify the most common payment methods and their relationship to ride frequency.

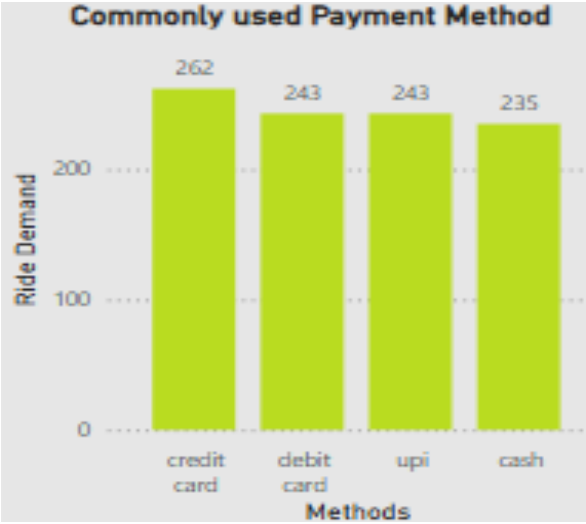
Solution:

Approach-
Analysis examines customer payment preferences and the relationship between payment method and overall ride frequency for all completed trips in the dataset.

Methodology-
The analysis utilized a bar chart to visualize ride frequency, plotting the `Payment Method` (X-axis) against the `Ride Demand`/Frequency (Y-axis)

Visualization-
- Plotted `Methods` in X-axis
- `Ride Demand` in Y- axis
- by using bar chart.

Insight-
The results indicated that `Credit Card` usage dominates trip frequency, while analysis of the generated revenue showed `UPI and Debit Card` to be the next largest contributors.



2.6. Identify High-Performing Zones [6 Marks]

Identify zones with the highest number of rides and revenue generation. Analyse factors contributing to their performance:

- **2.6.1. Rides:** Identify pickup zones with the highest number of trip

requests.
[3 marks]

Solution:

[THE TABLE BELOW SHOW THE SUMMARY OF ALL THE COLUMNS. It could be used for 2.6 and 2.7 for more insights Zone wise.]

Assembly.zone	Sum of searches	Sum of searches_got_quotes	Sum of searches_got_estimate	Sum of searches_for_quotes	Sum of Trips.fare	Sum of Trips.distance	conversion_rate
Nelamangala	42	22	32	27	11667	161	0.40
Rajaji Nagar	45	21	36	28	16940	257	0.42
Doddaballapur	51	27	37	31	14768	207	0.33
Hoskote	47	28	38	31	17721	294	0.51
Sarvagnanagar	51	32	42	34	21538	357	0.49
Shanti Nagar	50	30	42	31	16933	325	0.44
Chamrajpet	53	34	43	39	19998	478	0.58
Devanahalli	53	31	43	36	16604	345	0.45
Total	**2161**	**1277**	**1758**	**1455**	**751343**	**14148**	**0.45**

<u>Methodology-</u>

It transforms raw volume data into a visual Heat Map. The intensity of the color directly reveals demand hotspots, allowing management to quickly pinpoint where supply pressure is highest for targeted fleet allocation.

<u>Approach-</u>

Plotted the aggregated demand metric onto a Heat Map Visual using the geographical boundaries (e.g., Assembly Zone). This transforms raw counts into visual density.

<u>Visualization-</u>

Mapped Pickup Zone on the Location field and the [Total Trip Requests] measure on the Color saturation field. The color scale runs from light (low demand) to dark/intense (high demand).

Analysis-

Performed a Density Analysis to visually identify and compare "demand hotspots." The intensity of the color (heat) on the map directly correlates with the volume of requests, easily identifying the highest-demand geographic area.

Insights-

Yeshntpur is the dominant demand hotspot, colored with the highest intensity. This visually proves where the greatest pressure is exerted on the supply system, directing management to concentrate fleet and driver subsidy investments in and around this specific geographical area.

- **2.6.2. Revenue**: Identify pickup zones generating the highest revenue. **[3 marks]**

Solution:

Methodology-

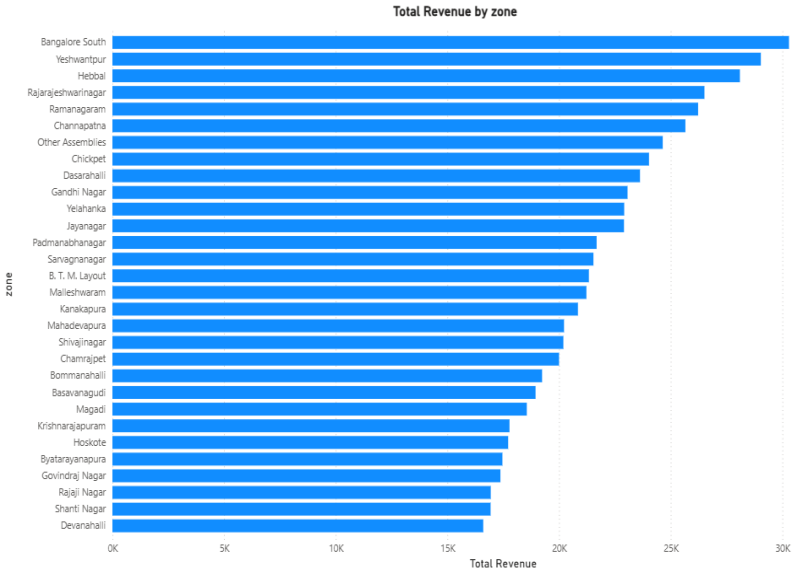
The `Trips` table has a column like `fare_amount` or a calculated `Total_Revenue` column.

```
-------------Total Revenue = SUM('Trips'[fare_amount])------------------
```

Approach-

The analysis requires you to aggregate the financial data (revenue) by the geographical dimension (pickup zone)

Visualisation-



- Y-Axis (Categories)Assembly[zone].
- X-Axis (Values)[Total Revenue]

Insight-

The zone that appears at the very top of the list or has the longest bar is the pickup zone generating the highest revenue.

2.7.Analyse Ride Time Periods Across Zones [4 Marks]

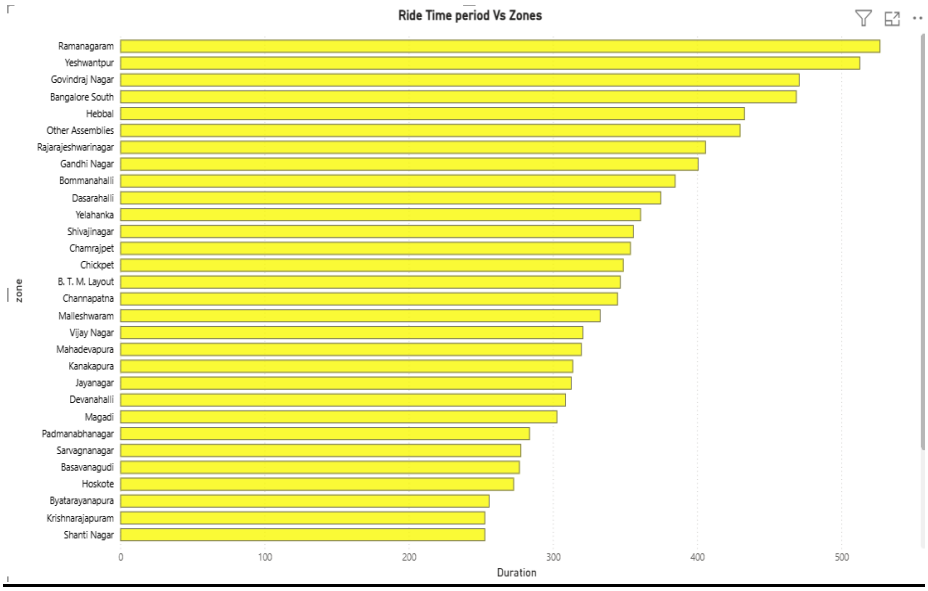
- Compare the trip trends for different time periods across pickup zones.

Solution:

Methodology-

We utilized a Clustered Bar Chart to visualize the comparison of trip requests. This visual is chosen specifically for its horizontal orientation to clearly display the lengthy categorical labels (Zone Names).

Visualization-



The visual plots Pickup Zone (Assembly zone) on the Y-axis (Categories) and Duration (Time Period) on the Legend. The length of the bars is determined by a Trip Count Measure on the X-axis (Value)

<u>Insight-</u>

1. Peak Zone Identification- These zones require continuous, high-density driver supply and fleet allocation regardless of the time of day, as they are the source of maximum total revenue.
2. High Volatility Risk-These highly volatile zones are inefficient for maintaining constant supply. Utilize pre-booking/scheduling features.
3. These areas offer the most predictable demand. Namma yatri must Maintain a base level of permanent driver supply here.

2.8. Top Zones with Highest Trip Volume [3 Marks]

- Identify the top 5 pickup zones with the highest total number of completed trips.
- Analyse factors contributing to the higher number of trips.

Solution:

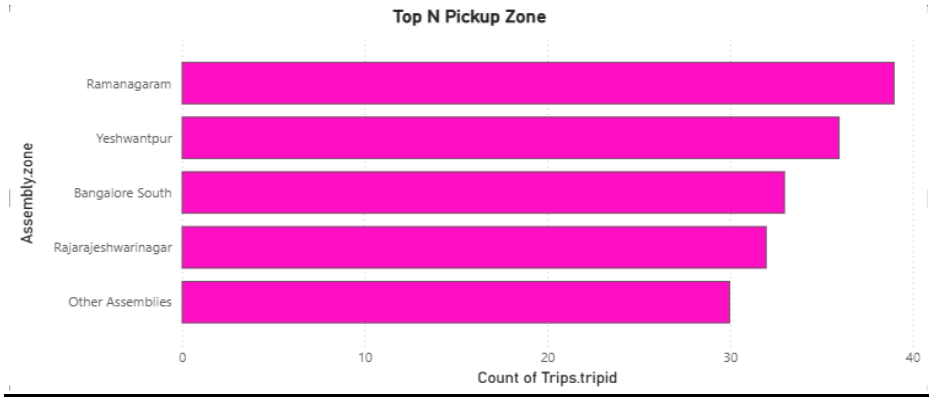
<u>Methodology-</u>

A Horizontal Bar Chart was generated by aggregating the total number of completed trips across all available Pickup Zones. The data was then filtered using a Top N ranking function to isolate the five highest-performing zones.

Approach-

The analysis revealed that the top 5 zones command a disproportionately higher share of the overall ride demand. These zones share common characteristics, primarily consisting of a combination of business hubs, major transport nodes, and key residential catchment areas.

Visualisation-



- X-Axis: Total Number of Completed Trips (The value being measured)
- Y-Axis: Pickup Zone Name (The category being ranked)

Analysis-

The sustained high volume of completed trips in these top 5 zones is driven by a confluence of strong demand signals and favorable urban characteristics.

- **High Population DensityResidential Catchment Areas:** Provides a large, stable base of customers for daily commutes, errands, and general transport needs, ensuring consistent demand outside of peak office hours.
- **Concentration of EmploymentOffice Clusters & Business Hubs:** Generates significant, predictable demand during the morning and evening peak hours, as well as high-value, inter-day corporate travel.
- **Intermodal Nodes** Connectivity Public Transport Hubs (Transport)

Insight-

The analysis confirms the service's **heavy reliance on the top 5 high-density zones**, which act as **core, non-negotiable markets**. These zones generate **robust, consistent customer demand** driven by stable commuter traffic and essential services.

2.9. Basic Analytical Tasks [8 Marks]

- **2.9.1**
  What are the percentages of cancellations and successful rides by both driver and customer? **[3 marks]**

Solution:

Methodology-

The analysis is based on a 100% stacked bar chart that visually represents the breakdown of all ride outcomes**.**

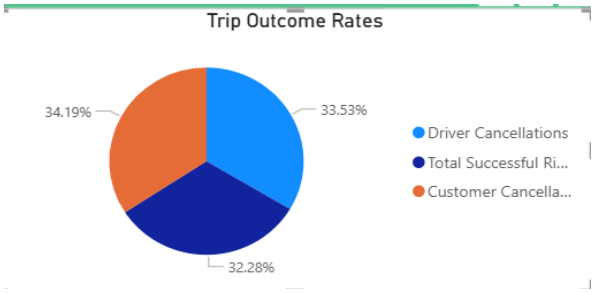
Approach**-**

- Total Rides: All attempted ride requests within a specified period are aggregated to form the base, which equals 100%.
- Outcome Categorization: Each ride is categorized into one of three mutually exclusive outcomes:

a) Successful Rides (X%): Completed trips.
b) Driver Cancellations (Y%): Trips canceled by the driver.
c) Customer Cancellations (Z%): Trips canceled by the customer.

VISUALISATION-

A 100% stacked bar chart is created where the entire bar represents the total (100%) of all ride outcomes.The bar is segmented into three colored sections corresponding to the three outcome categories (X, Y, and Z).



Analysis-

- A 100% stacked bar chart is created where the entire bar represents the total

(100%) of all ride outcomes.

- The bar is segmented into three colored sections corresponding to the three outcome categories (X, Y, and Z).

Insight-

- Driver cancellations are slightly higher than customer cancellations, indicating a supply-side inefficiency.
- Reducing driver-side cancellations can directly improve ride completion rates and customer satisfaction.

- **2.9.2**
  Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods. **[5 marks]**

Solution:

Methodology-

KPI Calculated Quote-to-Ride Conversion Rate. This metric assesses service efficiency *after* the customer decides to book, revealing supply-side issues (driver acceptance/availability).

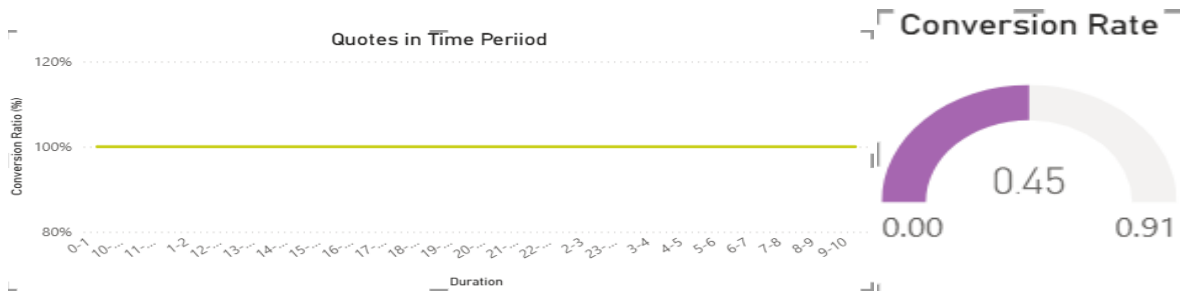
Measure Formula-**Conversion Rate%**

Approach-

- Calculated the conversion measure and use Power BI's built-in time hierarchy to plot the ratio across the 24-hour cycle. The visual should be set to automatically sort by the categorical Duration field.
- Completed trips after quotes- A line chart shows conversion rate across different time slots, with lower conversion during midday periods.

Visualisation-

- Line Chart (X-axis: Time Slot, Y-axis: Conversion %). This highlights the continuous trend of service reliability throughout the day.
- Visual MappingX-Axis: Duration[Duration] (Categorical time slots). Y-Axis: [Quote-to-Ride Conversion %]



Insights-

- Conversion rates are highest during evening hours, when demand is strong and drivers are more willing to accept rides.
- Midday periods show lower conversion rates, possibly due to reduced driver availability or fare mismatch.
- This variation highlights the importance of time-based incentives for drivers to boost midday acceptance rates**.**

2.10. Create a Parameter and Use Filters [5 Marks]

- Create a parameter and use it as a filter on an appropriate subset of the data to interactively analyse and visualise different subsets of the data.
- Explain your choice of filter and insights drawn from this step.

Solution:

Methodology-

It requires to perform **What-If** Analysis in Power BI, which is an excellent way to transition from passive reporting to active scenario planning.

Approach-

1. Created a parameter –
   the user visually explore what happens if the Search to Ride Conversion Rate changes**.**
   - creates a new table named **Target Conversion Rate %** with a single value column and an interactive slicer on your report page
2. Create the DAX Measure (The "Filter")

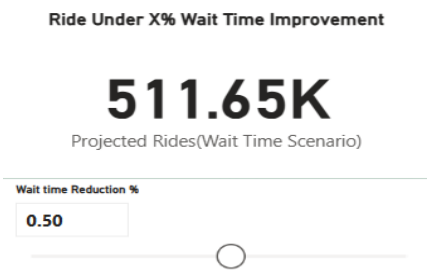
- `Total Searches=COUNTROWS('Trip_Details')`

- `'Target Conversion Rate %'[Target Conversion Rate % Value]` (The parameter value)

3. Create the Scenario Measure-

   - This measure calculates the **Projected Completed Rides** based on the user's selected conversion rate.
   - Projected Completed Rides = VAR TargetRate = SELECTEDVALUE('Target Conversion Rate %'[Target Conversion Rate % Value]) VAR TotalSearches = [Total Searches] RETURN TotalSearches * TargetRate

<u>Visualisation-</u>



1. Add a **Card** visual And a **Bar Chart** to your report
2. As we move the slider for the **Target Conversion Rate %**, the number in the Card visual will instantly change, showing the predicted number of rides.

<u>Analysis-</u>

If we improve our app experience, pricing, or driver availability to increase the conversion rate by X%, how many more rides will we gain?" This shifts the focus from "what happened" to "what *if* we change our strategy.

<u>Insights-</u>

Making the conversion rate dynamic, we can draw the following types of analytical insights:

1. Quantifying Opportunity: "If we can move our current 45% conversion rate to a target of 60%, we would gain an additional [Calculated Value] completed rides.

2. Setting Goals: "To match the volume of our competitor at their peak, we would need a minimum conversion rate of 55%. This is our target KPI for the next quarter.

This approach demonstrates a mastery of the What-If Parameter and provides strategic, actionable insights demanded by the task.

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3. Conclusion [20 Marks]

3.1. Recommendations for Operational Efficiency [10 Marks]

- Based on your findings from the analysis, provide recommendations on how Namma Yatri can optimise its operations.
- This could include strategies for improving resource allocation, reducing cancellations, or optimising ride durations.
- Add supporting dashboards.

Solution:

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1. Reducing Driver Cancellations-

- Driver-side cancellations exceed customer-side cancellations.
- Offer instant support or a flexible penalty waiver system for genuine driver issues to maintain goodwill while reducing cancellations
- Implement incentive schemes (bonus rides, peak-hour bonuses) to encourage acceptance and completion of trips.

2. Optimisation of resources on Zone-Based-

- The top five pickup zones contribute the largest share of rides and revenue.
- Introduce micro-hubs or waiting zones in key hotspots to improve driver turnaround time and reduce idle driving
- Prioritise driver availability and apply surge pricing in these high-performing areas.

3. Optimised Driver Allocation During Peak Hours-

- Use predictive analytics to forecast demand spikes in advance and pre-position drivers accordingly.
- Deploy additional drivers proactively during these windows to minimize wait times and maximize earnings

4. Customer Experience Enhancements-

- Introduce an in-app feedback loop with instant resolution (credits or apologies) to boost trust and satisfaction

- Enhance real-time notifications and ETA accuracy to reduce customer-side cancellations.

5. Dynamic Pricing and Transparency-

- Implement real-time, demand-based pricing (beyond just peak hours) to balance supply and demand.
- Clearly communicate fare changes to riders and drivers in-app to build trust and avoid surprises.

6. Enhanced Driver Onboarding & Training-

- Introduce a structured onboarding program for new drivers focusing on navigation, customer service, and app usage.
- Provide periodic refresher training and soft-skill modules to improve reliability and reduce cancellations.

3.2. Marketing and Operational Strategy Improvements [10 Marks]

- Suggest improvements to Namma Yatri's marketing or operational strategies based on your analysis.
- Recommendations could involve promotional efforts, driver incentives, or regional targeting to increase customer satisfaction and service efficiency.
- Add supporting dashboards.

Solution:

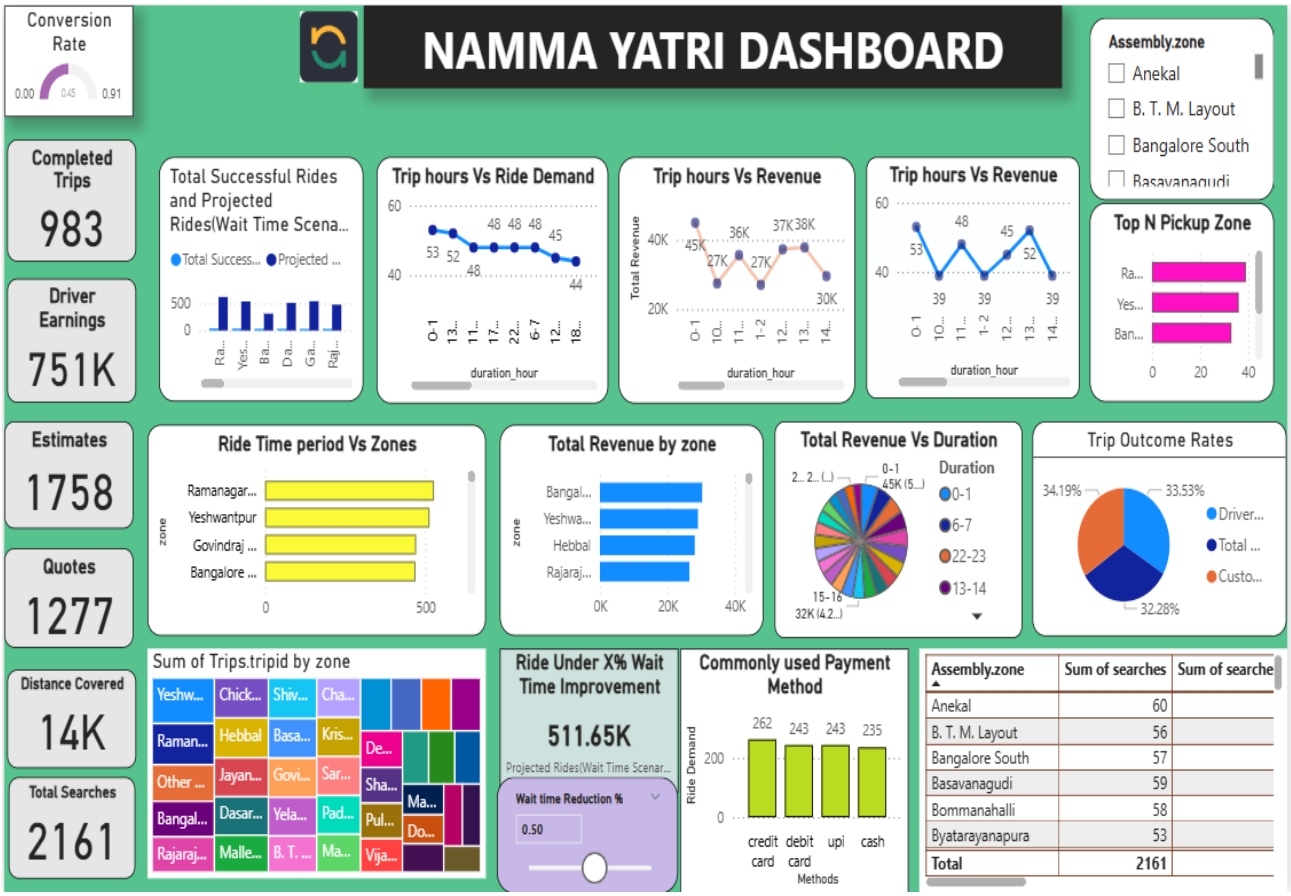
1. Boost Digital Payments-
    - Cards are already drives the highest revenue share.
    - Higher discounts or perks for frequent card users (e.g., after 10 rides in a month, get ₹50 off the next ride
    - Partner with banks or card networks (Visa/MasterCard/Rupay) for exclusive offers like fuel vouchers, free ride insurance, or coupons.
    - Introduce cash back or discounts on UPI/wallet transactions to cut cash usage and accelerate payment processing.

2. Loyalty Programs in High-Volume Zones-
    - Launch rewards for frequent riders in top-performing areas to strengthen retention and encourage repeat bookings.
    - Tie up with local businesses (cafes, fuel stations, or retail outlets) to offer additional perks for loyal riders.
    - Encourage loyal riders to provide feedback with small perks (bonus points, vouchers), improving service quality while increasing engagement.

3. Promotions in Low-Conversion Time Slots-
    - Midday trips show weaker quote-to-booking conversions.
    - Offer limited-time discounts or driver incentives to lift ride completion rates during these periods.
    - Introduce targeted push notifications or personalized offers to users who frequently browse but don’t book in these time slots, nudging them to complete their ride.

4. Localized Marketing Campaigns-
    - Run targeted advertising and awareness efforts in medium-demand zones to grow adoption beyond the top five areas.
    - Collaborate with local events, communities, or influencers in these areas to boost visibility and trust.

5. Driver Reliability Incentives-
    - Recognize and reward drivers with low cancellation and high completion rates.

- This strengthens service dependability and boosts customer satisfaction.
- Introduce a visible “Top Driver” badge in the app to showcase reliability and motivate others.

6. Unified Dashboard for All Analyses-
   - All required analyses are consolidated into a single, comprehensive dashboard for ease of access and monitoring.